PHYSICAL FORMULA

Ideal-gas entropy change

$$\Delta S = nC_v \ln\frac{T_f}{T_i} + nR \ln\frac{V_f}{V_i}$$

GENERATED QUESTION

A piston-cylinder device contains $1.0\,\text{mol}$ of a diatomic ideal gas. Initially, the gas is at $300\,\text{K}$ and occupies a volume of $0.01\,\text{m}^3$. After a process, the gas is heated to $400\,\text{K}$ and expanded to a volume of $0.02\,\text{m}^3$. Calculate the change in entropy for the gas, given that the molar heat capacity at constant volume is $20.785\,\text{J}\,\text{mol}^{-1}\,\text{K}^{-1}$ and the ideal gas constant is $8.314\,\text{J}\,\text{mol}^{-1}\,\text{K}^{-1}$. The answer is expressed in the unit $\text{J}\,\text{K}^{-1}$.